

Desca Affajry Rais

Mathematics Undergraduate · Researcher · Educator

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Rembang Regency, Central Java, Indonesia



PROFESSIONAL SUMMARY

I am a final-year Mathematics undergraduate at Universitas Diponegoro with a 3.92/4.00 GPA. My primary interests are pure mathematics, especially real and complex analysis, graph theory, abstract algebra, topology, and combinatorics. I also have experience in university mathematics teaching, olympiad coaching, academic-content development, data analysis, and computational projects.

EDUCATION

Sep 2023 — Present

ONGOING

Universitas Diponegoro

B.Sc. in Mathematics · GPA 3.92 / 4.00

Chair, UKM KOMIPA UNDIP (2025); teaching and tutorial experience in Real Analysis II; active in the Mathematics Student Association; ON-MIPA delegate and mathematics competition awardee.

Jun 2020 — Mar 2023

COMPLETED

SMA Negeri 1 Sulang

Mathematics and Natural Sciences · Final Grade 86.55 / 100.00

Earned regional and national mathematics awards, represented the school in OSN Informatics, and provided peer mathematics mentoring.

2026	DRAFT Distance Spectra of Corona Graphs of Complete Graphs and Paths via Invariant Subspaces Desca Affajry Rais · Suryoto, S.Si., M.Si. Studies the distance spectrum of $K_n \odot P_m$ using block matrices, Kronecker products, and invariant subspaces.
2026	DRAFT Connectivity of Int-Graphic Topology on Directed Graphs Desca Affajry Rais · YD Sumanto · Solikhin Characterizes connectivity of finite int-graphic Alexandroff spaces through minimal open neighborhoods and the out-neighborhood inclusion preorder.
2026	PRESENTED Modular Total Edge Irregular Labeling of Corona Product of Cycle C_4 with Path and Null Graphs The 16th ISNPINSA · Oral Presentation Developed explicit labeling constructions and established exact modular total edge irregularity strength results.
2026	COMPLETED Inequality Problem Solving Book Desca Affajry Rais · Drs. YD Sumanto, M.Si. · First Edition · 174 pages Proof-oriented mathematics book covering inequality theory, rigorous proofs, worked examples, and exercises. The manuscript is complete; Indonesian Intellectual Property (HKI) registration is planned.

PROFESSIONAL EXPERIENCE

Aug 2026 — Present

ONGOING

University Mathematics Tutor

Tutorkuliah.id

Tutoring in advanced undergraduate mathematics with emphasis on conceptual understanding and rigorous problem solving.

Jan 2026 — Present

ONGOING

Mathematics Olympiad Coach

KOMIPA UNDIP

University-level training in Real Analysis, Complex Analysis, Linear Algebra, Abstract Algebra, and Combinatorics.

Apr 2026 — Jun 2026

COMPLETED

Teaching Assistant

Asosiasi Matematika Terapan UNDIP

Real Analysis tutorials, proof-oriented problem solving, and academic consultation.

Dec 2025 — Mar 2026

COMPLETED

Mathematics Item Writer

SainsIn

Authored quantitative-reasoning items, detailed solutions, instructional videos, and quality-control materials.

Nov 2025 — Jan 2026

COMPLETED

Data Analyst

Emergencyy Call

Data cleaning, SQL querying, Excel reporting, KPI dashboards, visualization, and workflow documentation.

Nov 2025 — Dec 2025

COMPLETED

Mathematics Item Writer

SNBT.ID

Authored high-difficulty Quantitative Knowledge and Mathematical Reasoning items with structured solutions.

May 2025

COMPLETED

Project-Based Virtual Intern · Data Scientist

ID/X Partners x Rakamin Academy

Built a credit-risk prediction pipeline using Logistic Regression and Random Forest.

Mar 2025

COMPLETED

Project-Based Virtual Intern · Big Data Analyst

PT Kimia Farma Tbk x Rakamin Academy

Built a BigQuery analytical table and a Looker Studio performance dashboard.

LEADERSHIP & ORGANIZATIONS

Jan 2025 — Dec 2025

COMPLETED

Chairman
Mathematics and Sciences Olympiad Club (KOMIPA UNDIP)

Led strategy, daily operations, olympiad mentoring, ON-MIPA delegation, and external representation.

Jan 2025 — Dec 2025

COMPLETED

Senior Staff, Education and Reasoning Department
Mathematics Student Association, Universitas Diponegoro

Delivered mathematics tutorials, educational content, and learning-material development.

Feb 2024 — Jan 2025

COMPLETED

Junior Staff, Education and Reasoning Department
Mathematics Student Association, Universitas Diponegoro

Coordinated Academic Delegation, mentoring for 100+ students, and received Best Junior Staff and Best Department Staff recognition.

Oct 2023 — Dec 2024

COMPLETED

Staff, Networking and Public Relations Department
KOMIPA UNDIP

Supported ON-MIPA selection, served as Mathematics Liaison Officer, and led UKM Roadshow 2024.

2020 — 2023

COMPLETED

Moderator, Mentor, and Board of Governor
Brainly

Progressed through multiple moderation and mentoring roles, including supervision of 29 apprentice moderators.

2020 — 2022

COMPLETED

Organizational Leadership
Pelajar Islam Indonesia (PII), Rembang

Served as Head of Organizational Development and later General Secretary of the Rembang Regional Board.

COMMITTEE & VOLUNTEER EXPERIENCE

Facilitator Division Coordinator / Koordinator Divisi Fasilitator
27th Undip’s Mathematics Competition · Jan 2025 — Feb 2026

Curriculum Division Staff / Staf Divisi Kurikulum
LKMM-TD FSM 2025 · Jan — Oct 2025

Event Division Staff / Staf Divisi Acara
Introduction, Gathering, & Adaptation Mathematics 2025 · Mar — Oct 2025

Event Division Staff / Staf Divisi Acara
Industrial Visit Mathematics 2025 · Mar — Aug 2025

Equipment Division Staff / Staf Divisi Perlengkapan
Math Action for Social and Education 2025 · Mar — Jun 2025

Public Relations Division Staff / Staf Divisi Humas
Leadership Training FSM 2025 · Mar — Apr 2025

Event Division Coordinator / Koordinator Divisi Acara
26th Undip’s Mathematics Competition · Mar 2024 — Feb 2025

Chief Executive / Ketua Pelaksana
UKM Roadshow 2024 · Jun — Sep 2024

Volunteer Teacher / Pengajar Sukarela
Mathematics Clinic 2024 · Aug 2024

Equipment Division Coordinator / Koordinator Divisi Perlengkapan
ON-MIPA UNDIP Selection 2024 · Oct 2023 — Feb 2024

Mathematics Liaison Officer / Liaison Officer Matematika
ON-MIPA Coaching Center 2024 · Feb 2024

SELECTED PROJECTS

Feb 2026	COMPLETED Crowd Optimization and Smart Routing Strategy at Dufan Ancol Operations Research · Mathematical Modeling Applied M/M/s queueing, Little’s Law, greedy optimization, and knapsack concepts; proposed 14,000 daily visitors, a <45-minute waiting-time target, and 75% maximum utilization.
Feb 2026	ONGOING DMath Learning Mathematics Education · EdTech Ongoing mathematics learning platform for structured materials, guided practice, problem banks, olympiad preparation, and learning tools.
Feb 2026	COMPLETED Bike Sharing Analytics Python · EDA · Streamlit Identified weekday demand peaks, weekend casual-user behavior, and environmental drivers through end-to-end exploratory analysis.
Dec 2025	COMPLETED Financial Anomaly Detection System Scikit-Learn · K-Means · Random Forest Combined unsupervised pattern discovery with supervised transaction classification.
May 2025	COMPLETED Credit Risk Prediction Analysis SMOTE · Logistic Regression · Random Forest Analyzed 466,285 observations and reported Random Forest performance of 99.70% accuracy, 99.94% precision, and 99.46% recall.
Mar 2025	COMPLETED Kimia Farma Business Performance Analysis BigQuery · SQL · Looker Studio Built a centralized analytical table and multidimensional performance dashboard.
Mar–Apr 2025	COMPLETED Investment Portfolio Optimization Markowitz · Efficient Frontier · Sharpe Ratio Analyzed portfolio allocation across KLBF.JK, HEAL.JK, and KAEF.JK using risk–return optimization.

Nov 2024 — Present

ONGOING

BSI Scholarship Inspirasi Batch 4

Academic and leadership scholarship with coaching, mentoring, professional workshops, and social-impact activities.

Academic and leadership scholarship with coaching, mentoring, professional workshops, and social-impact activities.

Feb — Jul 2026

COMPLETED

Coding Camp 2026 — Data Scientist

Coding Camp powered by DBS Foundation

Coding Camp powered by DBS Foundation

Mar — Jun 2026

COMPLETED

Microsoft Elevate Training Center

Data Science with Microsoft Fabric · Generative AI with Microsoft Azure

Data Science with Microsoft Fabric · Generative AI with Microsoft Azure

Jan — Mar 2026

COMPLETED

IDCamp 2025 — Intermediate Data Scientist

Indosat Ooredoo Hutchison

Indosat Ooredoo Hutchison

Oct — Dec 2025

COMPLETED

IDCamp 2025 — Beginner Data Scientist

Indosat Ooredoo Hutchison

Indosat Ooredoo Hutchison

CERTIFICATIONS

UNDIP SEU English Proficiency Test EPT 65 · TOEFL Prediction equivalent 549 · 17 Sep 2026	TOEFL ITP Prediction Test 590 / 677
AI Engineer Bootcamp / Bootcamp AI Engineer	Foundations of Artificial Intelligence / Belajar Dasar AI
Foundations of SQL / Belajar Dasar SQL	Computer Networking Fundamentals / Fundamental Jaringan Komputer
Data Analyst — Basic Level ITBOX · Mar 2025	Python Programming Essentials: A Beginner's Guide Rakamin Academy · Mar 2025
Classical Cryptography for Beginners Cyber Academy Indonesia · Jul 2024	Introduction to Information Security Cyber Academy Indonesia · Jul 2024
Student Leadership and Management Training — FSM 2024 BEM FSM UNDIP · Apr 2024	Introduction to Data Science with Python DQLab · Jan 2024
Python Fundamentals for Data Science DQLab · Jan 2024	Basic Student Management Skills Training / LKMM Pra Dasar Sep 2023
Python Programming for Beginners CODEPOLITAN · Credential Z7R8TDD · Jul 2024 — Jul 2027	

SKILLS

Mathematics & Theory Pure Mathematics · Real Analysis · Complex Analysis · Graph Theory · Abstract Algebra · Topology · Combinatorics · Mathematical Modeling · Formal Proof Writing	Technical & Computing Python · SQL · LaTeX · Scikit-Learn · Looker Studio · Microsoft Excel · Microsoft Fabric · Microsoft Azure
Data & Analytics Data Analysis · Exploratory Data Analysis · Machine Learning · Data Visualization · Statistical Analysis · Data Engineering	Academic & Teaching Research · Academic Writing · Mathematics Olympiad · Teaching · Curriculum Development · Assessment Design · Problem Solving

AWARDS & ACHIEVEMENTS

Silver Medalist / Medali Perak · Isaac Newton Competition 2026 · OLIMNUS × UIN Sunan Kalijaga
Jul 2026

Silver Medalist / Medali Perak · Future Science Olympiad · POSI
Jun 2026

Bronze Medalist / Medali Perunggu · Mathematical Analysis and Geometry Day XVII · Institut Teknologi Bandung
May 2026

Gold Medalist / Medali Emas · Cendekia Natural Science Competition
Apr 2026

1st Place / Juara 1 · ON-MIPA UNDIP Selection 2026
Apr 2026

1st Place / Juara 1 · Indonesia Science Competition (KSI)
Mar 2026

Gold Award · KTOM March 2026
Mar 2026

1st Place / Juara 1 · Indonesia Science Competition (InSC)
Feb 2026

4th Place · Honorable Mention 1 · MATICS Mathematics Olympiad · Hasanuddin University
Feb 2026

Silver Award · KTOM February 2026
Feb 2026

Bronze Medalist / Medali Perunggu · POSI Science Competition
Jan 2026

2nd Place / Juara 2 · Indonesian Youth Science Olympiad #6
Jan 2026

2nd Place / Juara 2 · National Science League #2
Jan 2026

Silver Medalist / Medali Perak · Einstein Science Competition 2026
Jan 2026

Gold Award · KTOM December 2025
Dec 2025

Special Honour Participant · International Youth Math Challenge
Dec 2025

1st Place / Juara 1 · LOSEMA X
Nov 2025

1st Place / Juara 1 · Olimnus Science 2025
Nov 2025

Silver Medalist / Medali Perak · Isaac Newton Competition 2025
Jul 2025

3rd Place / Juara 3 · ASOE 4.0
May 2025

4th Place · ON-MIPA UNDIP Selection 2025
Mar 2025

Finalist / Finalis · MATHLICIOUS 2022
Nov 2022

1st Place / Juara 1 · ALHAMBRA Mathematics Competition
Feb 2022

Bronze Medalist / Medali Perunggu · Insight Arjuna Competition
Oct 2021

Gold Medalist / Medali Emas · Dirgantara Science Competition
Oct 2021

Silver Medalist / Medali Perak · Indonesian Youth Science Competition
Oct 2021

Bronze Medalist / Medali Perunggu · Bimage Competition Season 2
Sep 2021

Silver Medalist / Medali Perak · POSI Science Competition
Sep 2021

Top 10 · National Calculus Olympiad Nusantara
Aug 2021